

Neurologic Features of Immunoglobulin G4-Related Disease



Mahmoud AbdelRazek, MD^a, John H. Stone, MD, MPH^{b,*}

KEYWORDS

- Immunoglobulin G4 • Meningeal inflammation • Multiorgan disease
- Orbital myositis • Orbital pseudo tumor • Hypophysitis

KEY POINTS

- IgG4-related disease (IgG4-RD) is an immune-mediated condition that can involve any organ in the body, including the central and peripheral nervous systems.
- The most common neurological features of IgG4-RD result from disease in the orbits, pachymeninges, cavernous sinus, and pituitary gland and stalk.
- Serum IgG4, often found in high concentrations in patients with IgG4-RD, is not believed to play a direct pathophysiological role in mediating tissue injury in most patients with this condition. Rather, the elevated serum IgG4 is believed to be a counter-regulatory mechanism.
- IgG4-RD typically responds well to either glucocorticoid therapy or B cell depletion. However, most patients flare during or after these treatments.

INTRODUCTION

Immunoglobulin (Ig) G4-related disease (IgG4-RD) has been recognized as a discrete, multiorgan condition only since 2003.¹ This immune-mediated disease can affect both the central and peripheral nervous systems, as well as nearly any other organ in the body. The most common neurologic features of IgG4-RD result from disease in the (1) orbits; (2) pachymeninges, including complications of pachymeningeal inflammation within the orbital apex and cavernous sinus; and (3) substance of the pituitary gland and stalk, leading to diabetes insipidus or hypopituitarism; as well as (4) a peripheral neuropathy that can involve either peripheral or cranial nerves. Disease affecting the brain parenchyma is rare but reported. The diagnosis of IgG4-RD is often challenging, requiring careful correlation of pathologic or radiologic findings with the patient's

Disclosure: Dr J.H. Stone has received research grants and performed consulting for Roche (Genentech) in the area of immunoglobulin G4-related disease.

^a Department of Neurology, Mayo clinic, 200 First Street. SouthWest Rochester, MN 55905, USA; ^b Rheumatology Unit, Massachusetts General Hospital, 55 Fruit Street, Boston, MA 02114, USA

* Corresponding author.

E-mail address: jhstone@mgh.harvard.edu

Rheum Dis Clin N Am 43 (2017) 621–631

<http://dx.doi.org/10.1016/j.rdc.2017.06.010>

0889-857X/17/© 2017 Elsevier Inc. All rights reserved.

rheumatic.theclinics.com

clinical features. IgG4-RD often mimics malignancy, infection, and other immune-mediated conditions.

IgG4-RD must also be distinguished from a growing number of neurologic diseases in which IgG4 autoantibodies are recognized to play a role in the pathophysiology. Querol and colleagues,² 2013, found that in a subset of subjects with chronic inflammatory demyelinating polyneuropathy, for example, pathogenic IgG4 autoantibodies directed against contactin-1, the contactin-1-associated protein-1(CASPR1), or the neurofascin-155 molecules seem to drive the disease.³ Similarly, beyond the central and peripheral nervous systems, the autoantibodies associated with subsets of pemphigus and idiopathic membranous nephropathy are also primarily of the IgG4 subclass. These diseases are not part of the IgG4-RD spectrum. IgG4-RD is not known to be associated with any specific pathogenic autoantibody identified to date. Rather, it is associated with the infiltration of IgG4+ plasma cells into tissues and, in many patients, dramatic elevations of polyclonal serum IgG4 concentrations. In some cases, nonspecific immune complexes involving IgG4 antibodies may contribute secondarily to tissue injury, but this is not believed to be the primary mechanism of immunologic injury.

GENERAL CLINICAL FEATURES AND COMMON PRESENTATIONS

The presentation of IgG4-RD is typically subacute. Many patients have symptoms and signs for months or even years before the diagnosis is established. True fevers are unusual, but constitutional symptoms such as fatigue and weight loss are common. Many patients have arthralgias and other musculoskeletal symptoms, particularly enthesopathy, but frank arthritis is atypical and suggests other diseases.

IgG4-RD often presents as a mass lesion, regardless of the organ or body area affected: submandibular gland, pancreas, kidney, orbit, pituitary gland, or meninges.⁴ Many patients have symptoms of allergy or atopy, for example, allergic rhinitis, nasal polyps, chronic sinusitis, nasal obstruction, and rhinorrhea. Mild to moderate peripheral eosinophilia and serum IgE concentration elevations that sometimes exceed 10 times the upper limit of normal are common. The basis for these atopic features is not understood fully, but processes inherent to IgG4-RD itself rather than to atopy per se seem to contribute to the eosinophilia and IgE elevation.^{5,6}

PATHOLOGIC FINDINGS

Both histopathology findings and immunostaining studies are important in establishing the diagnosis, but neither histopathologic nor immunostaining findings are diagnostic of IgG4-RD in and of themselves. Close pathologic mimickers of IgG4-RD include granulomatosis with polyangiitis, inflammatory myofibroblastic tumors, and others. Rigorous assessment of the clinical features, serologic findings, and radiologic manifestations in light of the pathologic evidence is crucial in patient evaluations.

The typical histopathological findings include a dense lymphoplasmacytic infiltrate enmeshed by an irregularly whorled, storiform fibrosis. Obliterative phlebitis and a mild to moderate tissue eosinophilia are also present in a high percentage of cases. The preponderance of plasma cells within a pathologic specimen, usually but not always greater than 40%, is IgG4-positive. Neither the number of IgG4+ plasma cells per high-power field nor the IgG4+ to IgG+ plasma cell ratio is specific for the diagnosis of IgG4-RD. Multiple other conditions can be associated with infiltration of large numbers of IgG4+ plasma cells, emphasizing the importance of integrating these data with information from other parts of the evaluation before coming to conclusions with regard to diagnosis.

PATHOPHYSIOLOGY

Multiple components of the immune system participate in the pathophysiology of IgG4-RD. As previously noted, the IgG4 molecule itself is not believed to play a primary role in IgG4-RD, albeit there may be some contribution of IgG4-containing immune complexes to tissue injury in some cases. On the other hand, interactions between B and T lymphocytes on several levels seem to be involved in the pathophysiology of IgG4-RD. The prevailing disease concept is that both a specific CD4+ cytotoxic T lymphocyte (CTL) and cells of the B lymphocyte lineage (B cells and plasmablasts) are central to the disease. Using this construct, the CD4+ T cells orchestrating the disease are sustained by continuous antigen-presentation by B cells and plasmablasts.⁷ Plasmablasts (CD19 + CD38 + CD27 + CD20-cells) are generally elevated in the peripheral blood of patients with IgG4-RD, even in the setting of a normal serum IgG4 concentration.⁸ Both CD4+ CTLs and plasmablasts are expanded in an oligoclonal fashion in the peripheral blood.

B cells (contained within germinal centers) and IgG4+ plasma cells are typically located within germinal centers and at the periphery of germinal centers, respectively. In contrast, CD4+ T cells are typically dispersed throughout localized IgG4-RD lesions. These cytotoxic T cells actively secrete granzyme B and perforin, cytotoxic products more commonly associated with CD8+ T cells. The CD4+ CTLs T cells also elaborate interleukin-1, interferon-gamma, and transforming growth factor-beta, all of which may contribute to the fibrosis so characteristic of this disease.

Another CD4+ T lymphocyte, the T follicular helper cell, interacts with B cells both within lymph nodes and in secondary and tertiary lymphoid organs within sites of disease to drive the class switch to IgG4.⁹ The ultimate purpose of this class switch may be as a counterbalancing measure to the primary immune response, but this is not known for certain.

SEROLOGY

Among patients with IgG4-RD, the range of serum IgG4 levels is extremely broad. Elevations as high as 40-fold to 50-fold above normal are observed, in contrast, approximately 30% of patients with histopathologically confirmed disease have normal serum IgG4 concentrations.¹⁰ Substantial elevations in serum IgG4 levels are not diagnostic of IgG4-RD in and of themselves but may be extremely important in suggesting the possibility of this diagnosis. The correlation between serum IgG4 concentrations and the need for additional treatment is only approximate, but high serum IgG4 concentrations at baseline identify patients who are unlikely to respond with a sustained remission to a glucocorticoid course of moderate duration.¹¹

IgG4 is distinctly different from other IgG subclasses in its ability to undergo dissociation at its hinge region, recombining with other half-antibodies to form bispecific molecules.¹² IgG4 is known to bind the fragment crystallizable (Fc) portion of other immunoglobulins poorly and to be an ineffective activator of the complement cascade under most circumstances. The role of IgG in IgG4-RD may be to serve as an antigen sink at sites of disease, binding loose antigen within inflammatory foci but not inciting further tissue injury.

CLINICAL AND RADIOLOGICAL PRESENTATIONS OF NEUROLOGIC IMMUNOGLOBULIN G4-RELATED DISEASE

Neurologic manifestations of IgG4-RD can be the presenting symptom of the disease.^{13–18} IgG4-RD can involve the substance of the dura matter (ie, pachymeninges),

pituitary gland and stalk, peripheral nerves, and (rarely) the brain parenchyma. Important neurologic complications also ensue from involvement of structures within the orbit and cavernous sinus.

Two major patterns of neurologic involvement by IgG4-RD are recognized. First, neurologic structures can be affected by direct infiltration of chronic IgG4-related inflammation into the substance of the different anatomic components of the central or peripheral nervous system. Second, neurologic tissues can be compressed by the mass effect of a nearby tissue affected by IgG4-RD.

MENINGEAL DISEASE

IgG4-RD is among the most common causes of nonmalignant meningeal inflammation.¹⁹ IgG4-RD should be considered in cases with unexplained radiographic evidence of pachymeningeal disease, particularly if hypertrophic or nodular. Meningeal IgG4-RD manifests mainly in the pachymeninges (ie, dura mater) of the brain or spinal cord.²⁰ In a single-center cohort of 125 IgG4-RD subjects, IgG4-related pachymeningitis occurred in 3 (2.4%) subjects.²¹

Symptoms of IgG4-related pachymeningitis depend on the location of the disease, and occur secondary to 1 or more of several mechanisms. First, traction of the dura may lead to headache, typically retro-orbital or frontal when the dura of the anterior and middle cranial fossa is involved. Alternatively, when the dura of the posterior cranial fossa is affected, headaches in the occipital or vertex regions may ensue. Second, meningeal inflammation may encase or compress a nearby structure, typically a cranial nerve, resulting in a cranial neuropathy. Third, blockage of cerebrospinal fluid (CSF) flow may occur, typically at the foramina of Luschka and Magendie, leading to hydrocephalus (**Fig. 1**). A sudden escalation in headache severity should raise a suspicion for the occurrence of hydrocephalus, particularly if there is a positional component to the headaches.

Headache, the most common symptom, can be chronic, intractable, and associated with referred pain to a variety of cranial and facial structures because of dural traction. IgG4-related pachymeningitis involving the dura around the cavernous sinus (**Fig. 2**) can present similarly to Tolosa-Hunt syndrome with proptosis, diplopia, and retrobulbar pain. Other middle cranial fossa dural lesions can present with visual acuity problems or visual field loss due to optic nerve and chiasmal compression.²² Pachymeningeal disease at the orbital apices can present with diplopia (**Fig. 3**).

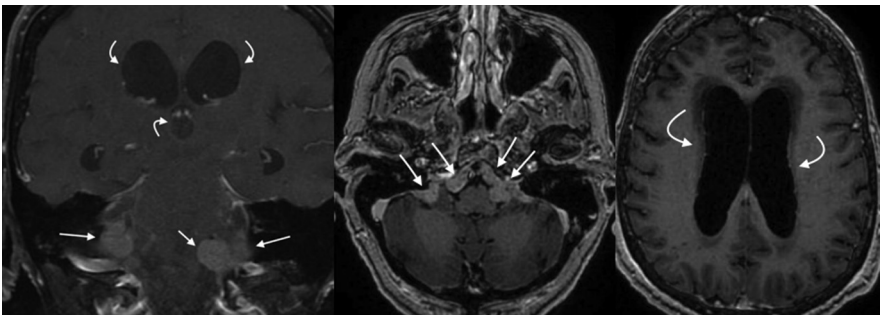


Fig. 1. A 71-year-old white woman with biopsy-proven IgG4-related pachymeningitis. This postcontrast T1 MRI shows ballooning of the lateral and third ventricles (*curved arrows*) likely secondary to obstruction of the foramina of Luschka and Magendie with pachymeningeal disease, seen here thickened and enhancing on MRI (*straight arrows*).

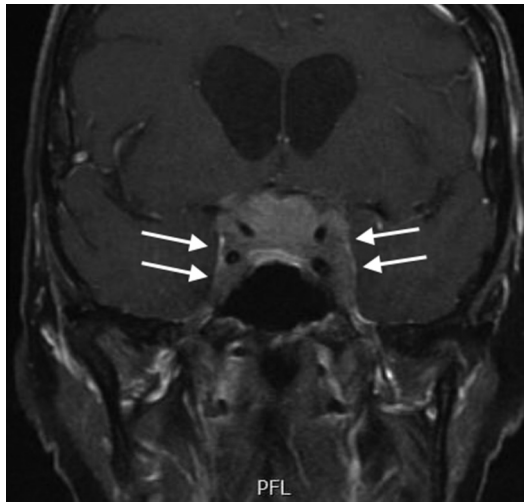


Fig. 2. Postcontrast T1 MRI demonstrating pachymeningeal disease (*arrows*) afflicting both cavernous sinuses.



Fig. 3. A 60-year-old white woman with a history of SLE and IgG4-related pachymeningitis diagnosed with meningeal biopsy. This T1-postcontrast MRI shows symmetric pachymeningeal thickening and enhancement (*arrows*) intracranially at the orbital apices, left greater than right. There was slight abduction deficit in the left eye without diplopia but no reduction of visual acuity.

Spinal IgG4-related pachymeningitis may occur in isolation, unaccompanied by other neurologic or non-neurological IgG4-RDs.^{14,15,23,24} It can manifest with a syndrome of myelopathy, radiculopathy, or myeloradiculopathy. Depending on location, this can present as neck, midback, or lower back pain that radiates to the limbs; progressive paraparesis or rarely quadriparesis; and/or bladder and bowel dysfunction.

IgG4-related pachymeningitis usually appears radiologically as homogenous dural lesions, diffuse thickening at a single site, or multiple discrete nodules. Computed tomography scans demonstrate IgG4-related pachymeningitis to be isodense compared with brain matter on noncontrast studies. MRI studies reveal these lesions to be hypointense or isointense on noncontrast T1-weighted and T2-weighted imaging.^{25,26} Both imaging modalities show homogenous enhancement on postcontrast studies, with lesions that enhance gradually on dynamic studies.^{25,26} Both adjacent bone destruction and intralesional calcifications are unusual in patients with IgG4-related pachymeningitis.²⁵

CEREBROSPINAL FLUID FINDINGS IN IMMUNOGLOBULIN G4-RELATED PACHYMENINGITIS

The CSF examination can be useful in the assessment of patients with potential IgG4-related pachymeningitis. Patients with IgG4-related pachymeningitis generally have a mild CSF pleocytosis with a lymphocytic predominance. In a series of 10 subjects, the mean (range) of nucleated cells within the CSF was 25/mm³ (22–260 mm³). The CSF protein is also frequently mildly elevated, with a mean (range) of 86 mg/dL (22–260) in the same series.²⁷ Oligoclonal bands can also occur in the CSF,²⁸ suggesting intrathecal production of IgG4. Larger numbers of subjects with IgG4-related pachymeningitis need to be studied in a systematic manner, however, before definitive conclusions can be drawn on this point.

One study compared the CSF findings of 3 subjects with IgG4-related pachymeningitis to those of 21 subjects with other causes of hypertrophic pachymeningitis, as well as to those of 9 controls. All 3 of the subjects with IgG4-related pachymeningitis had CSF IgG4 levels greater than 2.27 mg/dL. In contrast, only 5% of those with pachymeningitis of other causes and none of the controls had CSF concentrations this high. A cutoff of 0.47 for IgG4-Loc (an index for intrathecal IgG4 production) correctly categorized all 3 IgG4-related pachymeningitis subjects while misclassifying none of those in the other 2 groups.²⁸

ORBITAL DISEASE

Orbital structures that can be affected by IgG4-RD, leading to neurologic manifestations, include (1) the extraocular muscles and levator palpebrae; (2) the optic nerve (cranial nerve II) and ophthalmic artery; (3) the superior and inferior branches of the oculomotor nerve (III), trochlear (IV), frontal (V1), lacrimal (V1) and nasociliary (V1) nerves; and (4) the infraorbital (V2) and zygomatic (V2) nerves.

Inflammation associated with the disease can also infiltrate the extraocular muscles. Extraocular muscle enlargement is usually associated with dacryoadenitis.²⁹ The orbital soft tissue can also be involved (Fig. 4), in isolation or in association with other ocular structures, and typically presents with painless proptosis. Proptosis often occurs secondary to enlargement of any of the ocular adnexa with IgG4-related chronic inflammation, including the lacrimal glands, the extraocular muscles, or the orbital soft tissue itself. In addition, IgG4-RD can cause an orbital pseudotumor that impinges on or encircles the optic nerve. Orbital apex lesions afflicting the optic nerve and causing

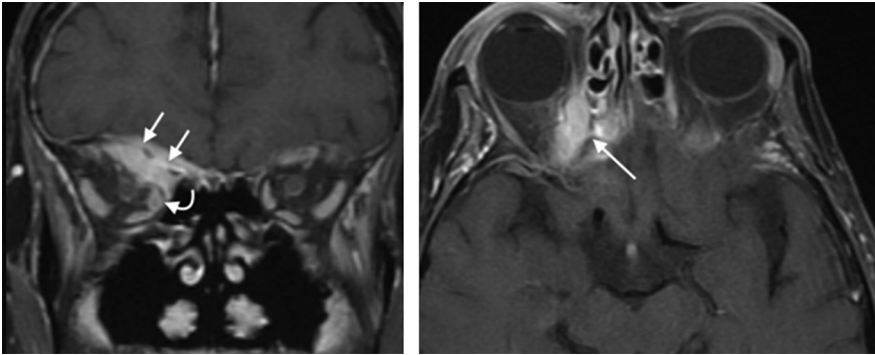


Fig. 4. Right eye orbital soft tissue IgG4-RD (*straight arrows*) displacing the right medial rectus (*curved arrow*) inferiorly.

reduced vision have been described, demonstrated radiologically as enlarged optic nerve sheaths.^{30,31}

Radiologic features of IgG4-related orbital disease include predominant involvement of the lateral rectus in terms of frequency and size of enlargement.³² In a review of 172 cases of IgG4-related ophthalmic disease, the structures most frequently involved on imaging were the lacrimal glands (62%–88%), orbital fat (29%–40%), extraocular muscles (19%–25%), and branches of the trigeminal nerve (10%–39%). Retro-orbital extension to involve the cavernous sinus was observed in 3 of 27 subjects in 1 series.³²

Inflammatory lesions of IgG4-related orbital disease are more frequently isointense or hypointense on T2 MRI sequencing compared with non-IgG4-RD causes of orbital infiltrative disorders.³³ This hypointensity likely reflects the hypercellularity and fibrosis that is characteristic of IgG4-RD pathologic findings.

PITUITARY GLAND AND STALK DISEASE

IgG4-RD commonly involves the pituitary gland and stalk in what is known collectively as IgG4-related hypophysitis.²⁰ This can be further classified according to the anatomic structure involved and its ensuing clinical picture into anterior gland disease, posterior gland involvement with stalk disease, and panhypophysitis. Panhypophysitis is the most common presentation of IgG4-RD in this area.³⁴ IgG4-related hypophysitis occurs commonly as a single-organ manifestation of the disease rather than in the context of multiorgan IgG4-RD.

Hormonal disturbances secondary to hypopituitarism are common in IgG4-related hypophysitis, perhaps more common than in lymphocytic hypophysitis.^{35,36} Hypopituitarism has been reported to occur in up to 83%^{34,35} of IgG4-related hypophysitis cases, diabetes insipidus in up to 72%,^{34,35} and both syndromes in 59% of patients.³⁴ Patients whose IgG4-RD primarily affects the anterior pituitary gland tend to present with reduced libido, hypogonadism, hypothyroidism, and hypoadrenalism. In contrast, patients with IgG4-RD involvement of the posterior pituitary gland or infundibular stalk present with central diabetes insipidus–related symptoms of polyuria and polydipsia.³⁷ Patients with panhypophysitis present with a combination of both syndromes. Constitutional symptoms such as headache, fever, weight loss, loss of appetite, lethargy, hypersomnia, and fatigue also occur.

Clinical outcomes are varied. Many patients do not regain pituitary function and require long-term hormonal replacement therapy despite radiographic resolution of

the pituitary lesions following treatment. Others, however, return to a functionally normal endocrine status after successful treatment.

MRI demonstrates an enlarged pituitary gland and/or thickened stalk with hypointense T2 lesions that homogenously enhances with gadolinium (**Fig. 5**).^{25,26} Cystic formations inside the enlarged gland can be seen.³⁴ Subsequently, an empty sella can develop after steroid therapy.³⁴ Loss of the normal posterior pituitary gland and stalk T1 precontrast hyperintensity is also well described.^{25,26,34}

PERIPHERAL NERVE DISEASE

Peripheral nerves can be involved with IgG4-RD by 1 of 2 mechanisms. First, the nerves can become compressed by the mass effect of local IgG4-RD involvement of non-neural tissue.³⁸ Second, the epineurium may be involved with chronic inflammatory aggregates caused by IgG4-RD, with or without further infiltration into the perineurium.^{30,39} Both mechanisms can result in varying degrees of axonal damage with ensuing loss of both large and small myelinated fibers,³⁹ with or without demyelination. There are no reports of inflammatory IgG4-RD cell infiltration into the endoneurium or nerve fibers on histopathology and thus this loss of axons is likely related to functional trophic changes as a result of chronic inflammation.

Branches of the trigeminal nerve, most characteristically the infraorbital nerve (**Fig. 6**), are perhaps the most frequently involved peripheral or cranial nerves. Pathologic sections of these involved nerves have demonstrated infiltration of IgG4-RD into the epineurium of the nerve.³⁰ The finding of infraorbital nerve involvement, however, is usually an incidental finding detected on the performance of head imaging for the evaluation of other IgG4-RD symptomatology. Similarly, asymptomatic spinal nerve root involvement was described in 3 subjects, none of whom had IgG4-RD involving any of the surrounding tissues.³⁰

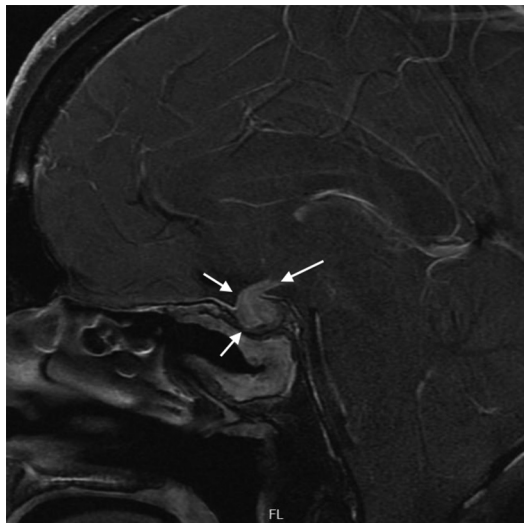


Fig. 5. T1 postcontrast MRI sequence demonstrating homogenous enhancement (arrows) of the anterior pituitary gland, posterior gland, and stalk.

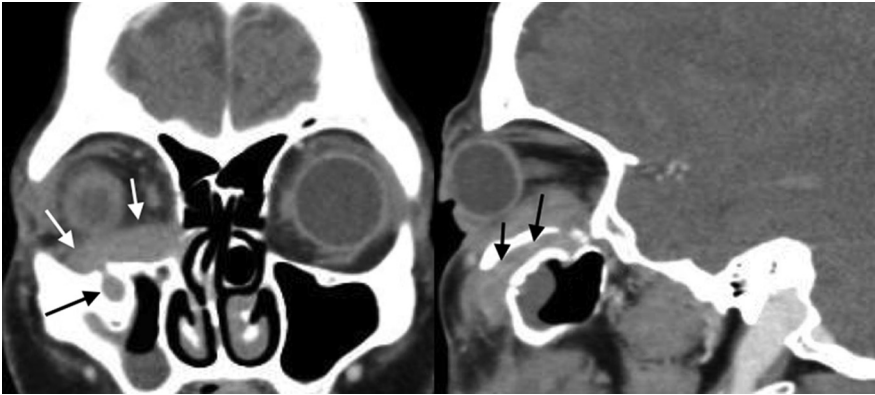


Fig. 6. IgG4-RD involving the right eye orbital soft tissue (*white arrows*) and the right infraorbital nerve (*black arrows*) that is thickened and expands the infraorbital canal, seen here on coronal (*right*) and sagittal (*left*) cuts.

CENTRAL NERVOUS SYSTEM PARENCHYMAL DISEASE

The substance of the brain and the spinal cord are rarely involved by IgG4-RD. The authors are aware of only 1 case with satisfactory evidence of brain parenchyma involvement.⁴⁰ Otherwise, there are 2 reports of IgG4-related pachymeningitis in which the inflammation extends to involve the underlying brain parenchyma.^{16,41} In those 2 cases, although the substance of the brain itself was not biopsied, the proximity of the parenchymal lesions and their concomitant resolution with their associated pachymeningitis after glucocorticoid therapy supports the notion of IgG4-RD pathologic complications. There is also a single account of an intraventricular IgG4-RD mass lesion with sufficient evidence in a 2009 case report,¹⁸ with no further cases of intraventricular IgG4-RD since that time.

TREATMENT

Data on the treatment of IgG4-related neurologic disease are sparse, consisting of only anecdotal reported experiences and small case series. Glucocorticoids are typically the first line of treatment of IgG4-RD, but these agents frequently fail to induce sustained disease remissions in the absence of (or even despite) ongoing treatment.⁴² For serious neurologic disease that threatens permanent neurologic injury or death, the authors recommend using rituximab in association with glucocorticoids, based on considerable experience with rituximab for the treatment of non-neurological manifestations of IgG4-RD.⁴³ It is worth noting, however, for the rare cases of IgG4-RD involving the brain parenchyma, that rituximab does not achieve good central nervous system concentrations in the presence of an intact blood-brain barrier.

REFERENCES

1. Kamisawa T, Funata N, Hayashi Y, et al. A new clinicopathological entity of IgG4-related autoimmune disease. *J Gastroenterol* 2003;38(10):982–4.
2. Querol L, Nogales-Gadea G, Rojas-Garcia R, et al. Antibodies to contactin-1 in chronic inflammatory demyelinating polyneuropathy. *Ann Neurol* 2013;73(3):370–80.

3. Querol L, Nogales-Gadea G, Rojas-Garcia R, et al. Neurofascin IgG4 antibodies in CIDP associate with disabling tremor and poor response to IVIg. *Neurology* 2014;82(10):879–86.
4. Kamisawa T, Zen Y, Pillai S, et al. IgG4-related disease. *Lancet* 2015;385(9976):1460–71.
5. Della Torre E, Mattoo H, Mahajan VS, et al. Prevalence of atopy, eosinophilia, and IgE elevation in IgG4-related disease. *Allergy* 2014;69(2):269–72.
6. Mattoo H, Della-Torre E, Mahajan VS, et al. Circulating Th2 memory cells in IgG4-related disease are restricted to a defined subset of subjects with atopy. *Allergy* 2014;69(3):399–402.
7. Mattoo H, Mahajan VS, Maehara T, et al. Clonal expansion of CD4(+) cytotoxic T lymphocytes in patients with IgG4-related disease. *J Allergy Clin Immunol* 2016;138(3):825–38.
8. Wallace ZS, Mattoo H, Carruthers M, et al. Plasmablasts as a biomarker for IgG4-related disease, independent of serum IgG4 concentrations. *Ann Rheum Dis* 2015;74(1):190–5.
9. Maehara T, Mattoo H, Ohta M, et al. Lesional CD4+ IFN-gamma+ cytotoxic T lymphocytes in IgG4-related dacryoadenitis and sialoadenitis. *Ann Rheum Dis* 2017;76(2):377–85.
10. Carruthers MN, Khosroshahi A, Augustin T, et al. The diagnostic utility of serum IgG4 concentrations in IgG4-related disease. *Ann Rheum Dis* 2015;74(1):14–8.
11. Wallace ZS, Mattoo H, Mahajan VS, et al. Predictors of disease relapse in IgG4-related disease following rituximab. *Rheumatology* 2016;55(6):1000–8.
12. Aalberse RC, Stapel SO, Schuurman J, et al. Immunoglobulin G4: an odd antibody. *Clin Exp Allergy* 2009;39(4):469–77.
13. Rice CM, Spencer T, Bunea G, et al. Intracranial spread of IgG4-related disease via skull base foramina. *Pract Neurol* 2016;16(3):240–2.
14. Radotra BD, Aggarwal A, Kapoor A, et al. An orphan disease: IgG4-related spinal pachymeningitis: report of 2 cases. *J Neurosurg Spine* 2016;25(6):790–4.
15. Gu R, Hao PY, Liu JB, et al. Cervicothoracic spinal cord compression caused by IgG4-related sclerosing pachymeningitis: a case report and literature review. *Eur Spine J* 2016;25(Suppl 1):147–51.
16. Li LF, Tse PY, Tsang FC, et al. IgG4-related hypertrophic pachymeningitis at the falx cerebrii with brain parenchymal invasion: a case report. *World Neurosurg* 2015;84(2):591.e7–10.
17. Takuma K, Kamisawa T, Igarashi Y. Autoimmune pancreatitis and IgG4-related sclerosing cholangitis. *Curr Opin Rheumatol* 2011;23(1):80–7.
18. Lui PC, Fan YS, Wong SS, et al. Inflammatory pseudotumors of the central nervous system. *Hum Pathol* 2009;40(11):1611–7.
19. Wallace ZS, Carruthers MN, Khosroshahi A, et al. IgG4-related disease and hypertrophic pachymeningitis. *Medicine* 2013;92(4):206–16.
20. Stone JH, Khosroshahi A, Deshpande V, et al. Recommendations for the nomenclature of IgG4-related disease and its individual organ system manifestations. *Arthritis Rheum* 2012;64(10):3061–7.
21. Wallace ZS, Deshpande V, Mattoo H, et al. IgG4-related disease: clinical and laboratory features in one hundred twenty-five patients. *Arthritis Rheumatol* 2015;67(9):2466–75.
22. Behbehani RS, Al-Nomas HS, Al-Herz AA, et al. Bilateral intracranial optic nerve and chiasmal involvement in IgG4-related disease. *J Neuroophthalmol* 2015;35(2):229–31.

23. Ezzeldin M, Shawagfeh A, Schnadig V, et al. Hypertrophic spinal pachymeningitis: idiopathic vs. IgG4-related. *J Neurol Sci* 2014;347(1–2):398–400.
24. Zwicker J, Michaud J, Torres C. IgG4-related disease presenting as dural thickening - a rare cause of myelopathy. *Can J Neurol Sci* 2014;41(3):392–6.
25. Katsura M, Mori H, Kunitatsu A, et al. Radiological features of IgG4-related disease in the head, neck, and brain. *Neuroradiology* 2012;54(8):873–82.
26. Toyoda K, Oba H, Kutomi K, et al. MR imaging of IgG4-related disease in the head and neck and brain. *AJNR Am J Neuroradiol* 2012;33(11):2136–9.
27. Lu LX, Della-Torre E, Stone JH, et al. IgG4-related hypertrophic pachymeningitis: clinical features, diagnostic criteria, and treatment. *JAMA Neurol* 2014;71(6):785–93.
28. Della-Torre E, Galli L, Franciotta D, et al. Diagnostic value of IgG4 Indices in IgG4-related hypertrophic pachymeningitis. *J Neuroimmunol* 2014;266(1–2):82–6.
29. Wallace ZS, Deshpande V, Stone JH. Ophthalmic manifestations of IgG4-related disease: single-center experience and literature review. *Semin Arthritis Rheum* 2014;43(6):806–17.
30. Inoue D, Zen Y, Sato Y, et al. IgG4-related perineural disease. *Int J Rheumatol* 2012;2012:401890.
31. Plaza JA, Garrity JA, Dogan A, et al. Orbital inflammation with IgG4-positive plasma cells: manifestation of IgG4 systemic disease. *Arch Ophthalmol* 2011;129(4):421–8.
32. Tiegs-Heiden CA, Eckel LJ, Hunt CH, et al. Immunoglobulin G4-related disease of the orbit: imaging features in 27 patients. *AJNR Am J Neuroradiol* 2014;35(7):1393–7.
33. Soussan JB, Deschamps R, Sadik JC, et al. Infraorbital nerve involvement on magnetic resonance imaging in European patients with IgG4-related ophthalmic disease: a specific sign. *Eur Radiol* 2016;27(4):1335–43.
34. Bando H, Iguchi G, Fukuoka H, et al. The prevalence of IgG4-related hypophysitis in 170 consecutive patients with hypopituitarism and/or central diabetes insipidus and review of the literature. *Eur J Endocrinol* 2014;170(2):161–72.
35. Caputo C, Bazargan A, McKelvie PA, et al. Hypophysitis due to IgG4-related disease responding to treatment with azathioprine: an alternative to corticosteroid therapy. *Pituitary* 2014;17(3):251–6.
36. Caturegli P, Lupi I, Landek-Salgado M, et al. Pituitary autoimmunity: 30 years later. *Autoimmun Rev* 2008;7(8):631–7.
37. Hattori Y, Tahara S, Ishii Y, et al. A case of IgG4-related hypophysitis without pituitary insufficiency. *J Clin Endocrinol Metab* 2013;98(5):1808–11.
38. Takahashi M, Shimizu T, Inajima T, et al. A case of localized IgG4-related thoracic periarteritis and recurrent nerve palsy. *Am J Med Sci* 2011;341(2):166–9.
39. Ohyama K, Koike H, Iijima M, et al. IgG4-related neuropathy: a case report. *JAMA Neurol* 2013;70(4):502–5.
40. Regev K, Nussbaum T, Cagnano E, et al. Central nervous system manifestation of IgG4-related disease. *JAMA Neurol* 2014;71(6):767–70.
41. Kim EH, Kim SH, Cho JM, et al. Immunoglobulin G4-related hypertrophic pachymeningitis involving cerebral parenchyma. *J Neurosurg* 2011;115(6):1242–7.
42. Khosroshahi A, Wallace ZS, Crowe JL, et al. International consensus guidance statement on the management and treatment of IgG4-related disease. *Arthritis Rheumatol* 2015;67(7):1688–99.
43. Carruthers MN, Topazian MD, Khosroshahi A, et al. Rituximab for IgG4-related disease: a prospective, open-label trial. *Ann Rheum Dis* 2015;74(6):1171–7.